MSPM0 USB-C controller board

A general-purpose USB-C powered MSPM0G3507 board I designed in KiCad, with a USB-to-UART bridge for talking to a PC and SWD for programming.

Rev A (schematic) pcb embedded usb-c kicad

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Author	Sao Aphisith Sithisack	Revision	Rev A (schematic)
Date	October 12, 2026	Tools	KiCad
Microcontroller	MSPM0G3507	Supply	3.3 V from USB-C
Status	schematic captured	Online	https://saosith.github.io/projects/mspm0-usb-c-controller-board/

Design only. Status: schematic complete in KiCad. Layout, fabrication and bring-up come next, so no values on this page are measured.
VBUS = 4.75 to 5.25 V (USB 2.0 range), room temperature

Abstract

This is a general-purpose microcontroller board for my projects. It runs from a laptop's USB-C port, appears to the PC as a serial port, and can be reprogrammed on the bench without extra hardware. The design centres on a TI MSPM0G3507 microcontroller with a CH340E USB-to-UART bridge, USB ESD protection, a 3.3 V LDO, an 8 MHz crystal and a Tag-Connect SWD footprint. Status: schematic complete in KiCad. Next: layout, fabrication and bring-up.

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Electrical characteristics

Parameter	Sym.	Target	Simulated	Unit	vs. target
3.3 V rail	V_3V3	3.3 ± 3%	–	V	—
Total 3.3 V load current [1]	I_load	≤ 100	–	mA	—
LDO dissipation at full load [2]	P_LDO	≤ 0.2	–	W	—
UART baud-rate error (115200)		< 1	–	%	—
ADC reference noise [3]		< 1 LSB (0.8 mV)	–		—
USB ESD, D+ and D– [4]		±8 kV contact	–		—

Conditions VBUS = 4.75 to 5.25 V (USB 2.0 range), room temperature

1 One USB 2.0 unit load, so the board works on any port before enumeration.

2 P = (5.25 V − 3.3 V) × 0.1 A ≈ 0.2 W, the worst case for the SPX3819 LDO.

3 12-bit ADC with a 3.3 V reference: 3.3 V / 4096 ≈ 0.8 mV per step.

4 IEC 61000-4-2 contact discharge, the level the USBLC6-2 array is rated for.

Introduction

I wanted a general-purpose microcontroller board I could reuse across projects. I wanted it to run off my laptop's USB-C port, send data to the PC over a serial port, and be reprogrammed on the bench without extra hardware. So I designed this board around a TI MSPM0G3507 (Arm Cortex-M0+) [1]. It has USB-C power, a USB-to-UART bridge, a crystal, an SWD debug pad and a filtered ADC reference.

Status: schematic complete. **Next:** layout, then fabrication and bring-up. No values on this page are measured.

Requirements

- **Power** from a laptop's USB-C port, with no separate bench supply.
- **Serial link** to a PC for logging, with no extra adapter.
- **Reprogramming** on the bench without extra hardware on the board.

- **Clean ADC reference**, so analog measurements are not limited by reference noise.

The table at the top lists the numeric targets. They come from the USB 2.0 limits and the part datasheets; nothing is measured yet.

System overview

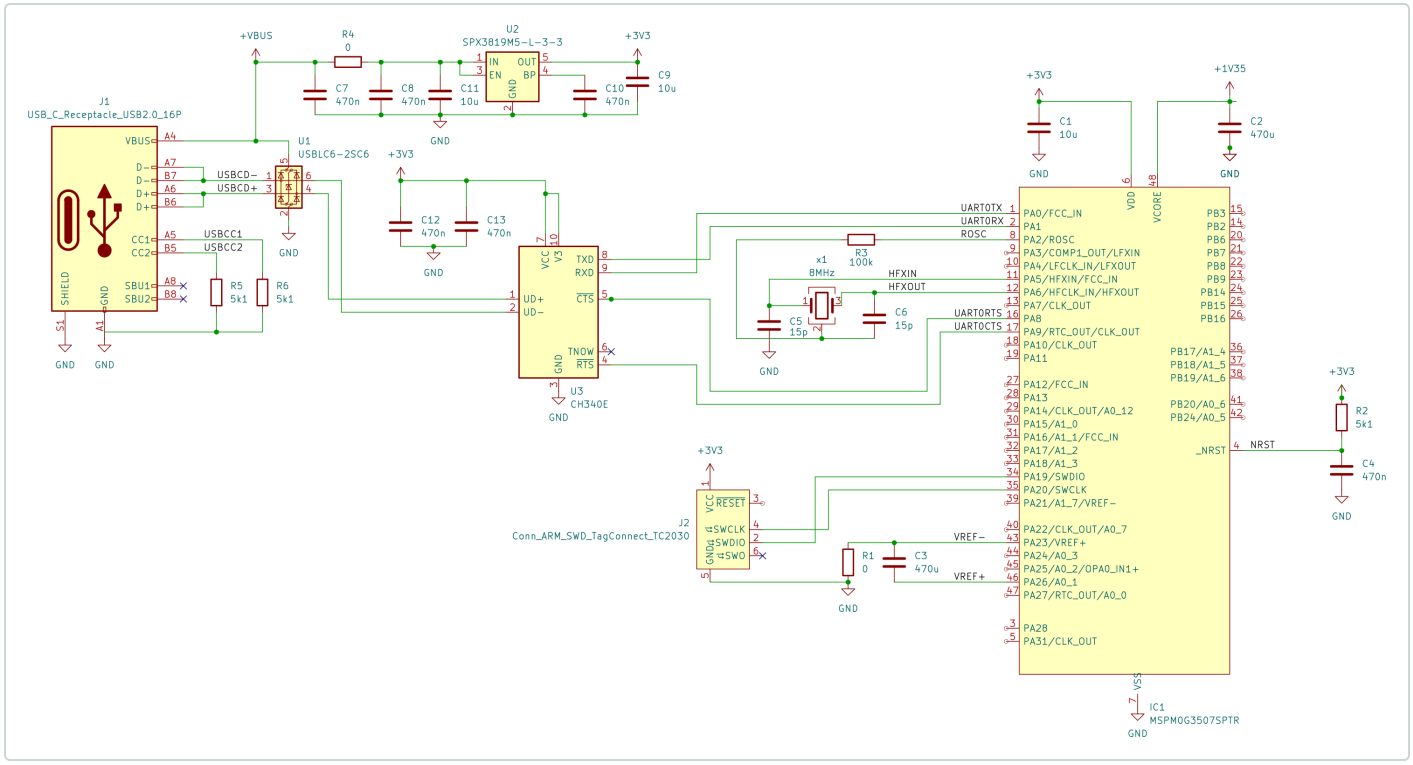


Fig. 1 The full schematic. Power comes in on the left, the microcontroller is on the right, and the USB-to-UART bridge is in the middle.

Design

Power path

USB-C supplies 5 V on VBUS. Normally, a USB-C host or charger keeps VBUS off until it detects a device. **R5 and R6 pull CC1 and CC2 to ground through 5.1 kΩ**, the value the USB Type-C specification sets for a sink [5], so the host sees the board as a device requesting power. VBUS then passes through R4, a 0 Ω link that can be replaced with a fuse

or removed to cut the supply. An SPX3819 low-dropout regulator (LDO) [4] drops the 5 V to 3.3 V and dissipates the difference as heat.

Data path and protection

D+ and D− pass through a USBLC6-2SC6 ESD array [3] first. If someone touches the connector, the electrostatic discharge (ESD) is clamped to ground before it reaches the CH340E USB-to-UART bridge [2]. In layout, I'll place the array right at the connector so the discharge has the shortest path to ground. The CH340E appears to the PC as a standard serial port and to the microcontroller as a two-wire UART.

Clock, reset, reference

An 8 MHz crystal with two 15 pF load capacitors drives HFXIN/HFXOUT. A crystal is typically accurate to tens of ppm, while an on-chip RC oscillator is usually off by around a percent, which is enough to disturb UART timing. The pull-up and capacitor on NRST hold the microcontroller in reset until the 3.3 V rail is stable. A 100 kΩ resistor on ROSC sets the reference current for the internal oscillator.

An RC low-pass filter sits on the ADC reference. The ADC measures every input as a fraction of this reference voltage, so noise on the reference appears in every reading.

Programming and debugging use a TC2030 Tag-Connect footprint: the cable's spring pins press onto pads on the board, so no header is soldered on.

Design decisions

Decision	Options considered	Choice	Why	Status
Clock source	Internal oscillator, or an external 8 MHz crystal	**8 MHz crystal (two 15 pF load caps)**	A crystal is typically accurate to tens of ppm, while an on-chip RC oscillator is usually off by around a percent. I chose the crystal so the UART baud rate and any timing-critical measurements do not depend on that error.	decided
Programming connector	Soldered pin header, or a TC2030 Tag-Connect footprint	**TC2030 Tag-Connect**	The Tag-Connect cable's spring pins press onto bare pads, so the board needs no header. I chose it to save board area and one part.	decided
Crystal load capacitors	15 pF as drawn, or a value recomputed from the crystal's rated load capacitance	**Not settled**	To verify before layout: C = 2 × (CL − Cstray), using the chosen crystal's rated load capacitance CL and an estimate of the board's stray capacitance.	open

Project status

Sections for layout, simulation and test results are added as each stage is completed.

Stage	Status	Plan
Schematic	✓ Complete	—
Design review	▶ In progress	Check decoupling, the serial handshake wiring and the ADC reference filter against the MSPM0 [1] and CH340E [2] datasheets.
Simulation	○ Planned	ADC reference filter response and LDO start-up.
Layout	○ Planned	Stack-up, placement and routing, with the ESD array right at the USB-C connector.
Fabrication	○ Planned	—
Bring-up and test	○ Planned	Rail voltages, USB enumeration, UART loop-back and SWD flashing.

Open issues

- **No auto-reset.** The bridge's handshake lines are not connected to NRST, so firmware loads through SWD and the PC cannot reset the microcontroller. Wiring RTS/DTR to NRST is an option for Rev B.
- **Design review before layout.** Check decoupling, the serial handshake wiring and the ADC reference filter against the MSPM0 [1] and CH340E [2] datasheets.
- **Crystal load capacitors.** Recalculate with C = 2 × (CL − Cstray) once the crystal part number is fixed.

Revision history

Rev	Date	What changed
A	`2026-09-10`	Schematic captured in KiCad and first written up.
A	`2026-10-12`	Write-up restructured as a report: abstract, decisions, BOM appendix, placeholders for layout and test.

References

1. Texas Instruments, MSPM0G350x mixed-signal microcontroller datasheet
2. WCH, CH340 USB-to-serial bridge datasheet
3. STMicroelectronics, USBLC6-2 ESD protection datasheet
4. MaxLinear, SPX3819 500 mA low-dropout regulator datasheet
5. USB Type-C Cable and Connector Specification (sink Rd pull-down on CC1/CC2)

Appendix: Bill of materials

Ref	Part	Value	Package	Source
IC1	MSPM0G3507SPTR microcontroller	—	LQFP-48 7x7 mm	LCSC C22362630
U2	SPX3819M5-L-3-3 LDO regulator	3.3 V	SOT-23-5	TBD
U3	CH340E USB-to-UART bridge	—	MSOP-10	TBD
U4	LIS2DH accelerometer	—	LGA-14 2x2 mm	TBD
D1	USBLC6-2SC6 ESD array	—	SOT-23-6	TBD
J1	USB-C receptacle (USB 2.0, 16 pin)	—	GCT USB4105 footprint	TBD
x1	Crystal	8 MHz	3225 4-pin	TBD
C1, C2, C4, C6, C7, C13	Capacitor	470 nF	0201	TBD
C3, C5, C11	Capacitor	10 µF	0805	TBD
C8, C9	Capacitor	15 pF	0201	TBD
C10, C12, C14	Capacitor	470 µF (as drawn)	0201	TBD
R1, R2, R6–R8	Resistor	5.1 kΩ	0201	TBD
R3, R4	Resistor	0 Ω	0201	TBD
R5	Resistor	100 kΩ	0201	TBD